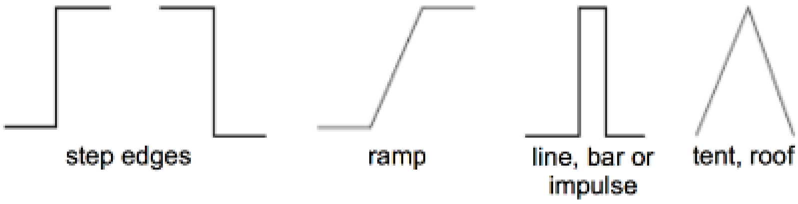
LAB Knowledge

1. Convolution of image with mask H*I where I is image and H is mask. Flip the mask so the columns and rows are swapped. Multiply over all overlapping pixels and sum to the centre pixel. Zero pad +where necessary.
2. Intensity level discontinuities can occur because of – depth, orientation, reflectance(surface material properties) and illumination(shadows) discontinuity.
3. Calculate gradient direction and orientation of matrix (edge detection lecture)
4. directional derivatives

   An edge is a location where the magnitude of the intensity gradient is high in the direction orthogonal to the edge
   An edge is an image region in which the grey level gradients have a common direction and large magnitudes

   Idealised Edge Types:

   The difference masks considered previously will detect these edges:

-1	1

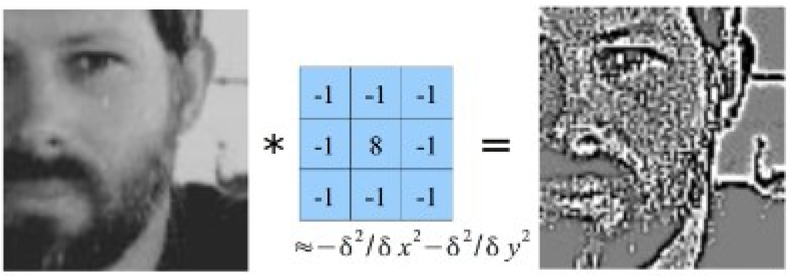
   $-\delta/\delta x$

-1	2	1

   $-\delta^2/\delta x^2$

   The Laplacian mask

   Can be seen as a combination of difference masks in each direction. It therefore detects intensity discontinuities at all orientations.

   Note: the Laplacian is actually the additive inverse of this mask.

   However, such a mask is most sensitive to a single dot, one pixel in size (i.e. noise).

5. Canny edge detector – based on Gaussian derivative masks
   a. Convolution with derivative of gaussians masks
   b. Calculation of magnitude $M = \sqrt{(I_x)^2+(I_y)^2}$ and direction $D=\arctan(I_y / I_x)$
   c. Non maximum suppression (thin multi-pixel wide ridges down to a single pixel by removing current pixel if neighbouring pixels perpendicular to the direction of the edge have a higher magnitude) used to find the bounding box or define the edge where magnitude is highest.
   d. Thresholding using a low and high threshold (accept all edge pixels with a magnitude over low threshold that are connected to a pixel with a magnitude over the high threshold)

6. Difference of Gaussian (DoG) mask is an approximation of the Laplacian of Gaussian mask (detects intensity discontinuities at all orientations), a mask which combines Gaussian smoothing with a Laplacian operator to detect edges while also supressing noise.
7. Image pyramid looks to find features with invariance to scale. It presents masks that can be viewed as a template for an image. The mask is convolved with an image and features are detected where features convolved with the mask exceed the threshold. An image pyramid looks to apply the filter at different sizes to an image at a fixed size.
8. Increasing the scale of an image is achieved by down sampling. Nearest neighbour and bilinear interpolation are methods of down sampling used. The pyramid is created by iteratively sub sampling the previous sub sampled image.
9. Histograms used as features - many to one (many images give the same histogram as the pixels can be rearranged. Can be used in classification and recognition). Invariant to geometric image operations – rotation, scaling, mirroring and skew.
10. LBP descriptors (local binary patterns) – describes the surroundings of a pixel, generates a bitcode from the binary derivatives of a pixel. Based on the assumption that texture has locally two complementary aspects – a pattern and its strength. For the 8 pixels surrounding the centre pixel C create a 8 bit digit. If bi>C then bi =1 else bi=0. Drawbacks of LBP is that you can miss important features like edges.
11. Invariance to illumination, scale, rotation, affine and perspective projection
12. Extracting SIFT features
    a. Determine approximate location and scale of keypoints (look for intensity changes using difference of gausians at two nearby scales)
    b. Refine location and scale
    c. Determine orientations for each keypoint (compute gradient magntiudes and orientations in a small window around the keypoint at the appropriate scale)
    d. Determine descriptors for each keypoint
13. Similarity measure - sum of absolute differences is the measure of the similarity between image blocks calculated by summing the L1 norm difference between image blocks
14. Region growing starts with a single pixel and labels all unlabelled neighbouring pixels with the same label if within a similarity threshold. Repeat until region stops growing and then start again with another seed pixel
15. Region merging starts with each pixel as its own region continue merging until regions cannot be merged anymore

Exam knowledge